CREDIT UNDERWRITING

& PORTFOLIO RISK MEMORANDUM

IFRS 9 Retail Credit Portfolio

Auto Loans | Credit Cards | Personal Loans

$129.46M	$2.919M	2.255%	14,970
Gross Exposure	Weighted ECL	Allowance Rate	Reporting Accounts

Committee view

ACCEPTABLE / MAINTAIN WITH ENHANCED MONITORING. Portfolio-wide credit risk is controlled, but Stage 2 loss concentration, Credit Card severity, and Auto LGD production support require explicit watchlist and model-risk monitoring.

Prepared for

Credit / Risk Committee

Reporting date: 31 December 2025

Portfolio data: synthetic | Models: SAS production + Python challenger

Purpose

A portfolio-style underwriting memorandum demonstrating how credit analysis, model validation, IFRS 9 staging, lifetime ECL, production controls, and management reporting can be integrated into one committee package.

01 | EXECUTIVE DECISION

Credit Committee Executive Summary

Portfolio underwriting conclusion, principal risks, and proposed risk posture

5.01%	47.53%	+30.7%	5 / 5
Stage 2 Exposure	Stage 2 ECL Share	Downside vs Base	ECL Governance

Illustrative underwriting recommendation

MAINTAIN / APPROVE PORTFOLIO RISK POSITION WITH CONDITIONS. No evidence supports a portfolio-wide de-risking action, but incremental risk management is warranted for Stage 2 accounts, high-severity unsecured exposures, and Auto accounts affected by the LGD production guardrail.

Primary strengths

- Strong PIT PD discrimination: OOT AUC 0.897 with VALID-only calibration and no TEST tuning.
- LGD champion improves test error versus product baselines; EAD / CCF governance passes 9 of 9 product-horizon checks.
- Scenario ordering is economically coherent: Upside ECL < Base ECL < Downside ECL.
- Monitoring mart reconciles exactly to the frozen risk master; Python challenger reproduces aggregate ECL within +0.17%.

Primary risks / conditions

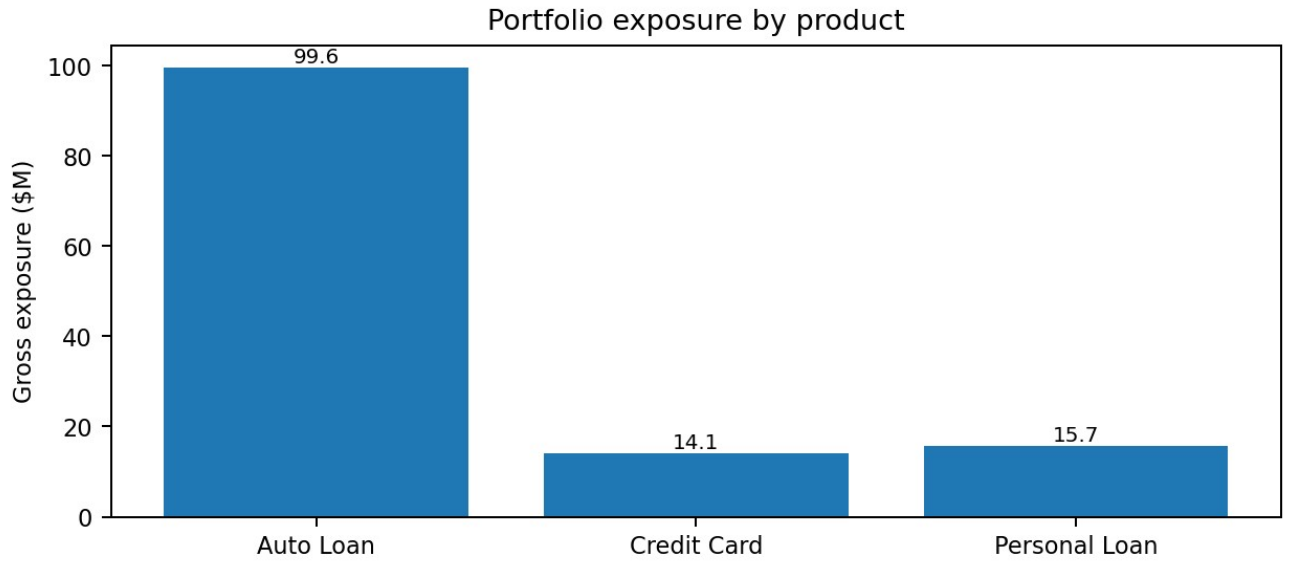
- Stage 2 represents only 5.01% of exposure but 47.53% of total ECL, creating disproportionate deterioration risk.
- Credit Card coverage is structurally high because unsecured LGD is severe; Stage 2 Card allowance rate is ~40.7%.
- Auto LGD production collateral coverage materially exceeds development support in selected accounts; TRAIN-P99 guardrail is required.
- Stage 3 is a reporting-date visible/default population rather than a fully reconstructed workout-stock ledger.

Underwriting note: Recommendation is illustrative and portfolio-oriented; it is not a real bank credit approval or regulatory conclusion.

02 | PORTFOLIO OVERVIEW

Portfolio Scope, Product Mix & Exposure

Synthetic retail credit book used for the underwriting and IFRS 9 assessment



Product	Accounts	Exposure	Weighted ECL	Allowance	Avg current PD
Auto Loan	5,220	$99.59M	$0.845M	0.85%	2.89%
Personal Loan	2,064	$15.74M	$0.696M	4.42%	4.80%
Credit Card	7,686	$14.13M	$1.378M	9.75%	4.39%

Underwriting read-through

Auto carries most of the balance exposure, while Credit Card drives the highest allowance intensity. The portfolio is therefore not dominated by one single risk dimension: secured balance concentration and unsecured severity must be managed separately.

03 | UNDERWRITING ASSESSMENT

Risk Assessment: Strengths, Weaknesses, Mitigants & Exceptions

Dimension	Assessment	Committee interpretation
Portfolio scale	$129.46M exposure / 14,970 accounts	Diversified at account level; concentration should be assessed by expected loss, not balance alone.
Credit deterioration	Stage 2 = 5.01% of exposure	Small balance share but outsized ECL contribution requires targeted monitoring.
Unsecured severity	Card / Personal LGD structurally high	Loss mitigation depends more on PD control and limit management than collateral recovery.
Secured severity	Auto collateral-sensitive LGD	Collateral coverage provides mitigation, but production extrapolation required a support guardrail.
Scenario risk	Downside +30.7% vs Base	Material but ordered stress response; portfolio is macro-sensitive without scenario inversion.
Model governance	5/5 ECL gates; 3/3 reporting gates	Control environment is strong for a portfolio project, with explicit limitations retained.

Credit strengths

- High-performing Stage 1 base: 94.59% of exposure remains performing.
- Time-based PD validation and fresh EAD FINALTEST reduce model selection bias.
- Scenario-weighted ECL and exact reporting reconciliation support auditability.

Credit weaknesses / exceptions

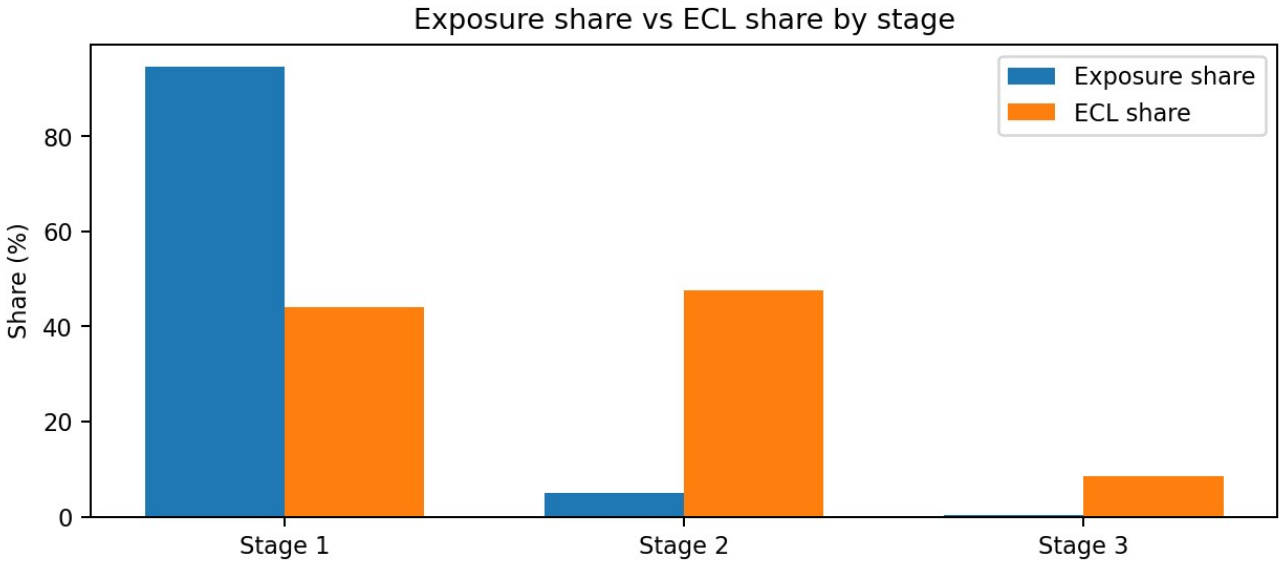
- Stage 2 is loss-concentrated relative to balance.
- Credit Card / Personal Loan recovery economics are weak by design and carry high LGD.
- Stage 3 workout accounting is simplified and should not be presented as a complete default-stock engine.

Risk posture

MODERATE / CONTROLLED. Maintain exposure subject to Stage 2 escalation, guardrail monitoring, scenario sensitivity review, and explicit Stage 3 limitation disclosure.

04 | CURRENT CREDIT QUALITY

IFRS 9 Stage Profile & Current Risk Posture



Stage	Accounts	Exposure	Exp. share	ECL	ECL share	Coverage
Stage 1 - Performing	13,658	$122.45M	94.59%	$1.287M	44.08%	1.05%
Stage 2 - SICR	1,251	$6.49M	5.01%	$1.387M	47.53%	21.39%
Stage 3 - Credit-impaired	61	$0.52M	0.40%	$0.245M	8.39%	46.89%

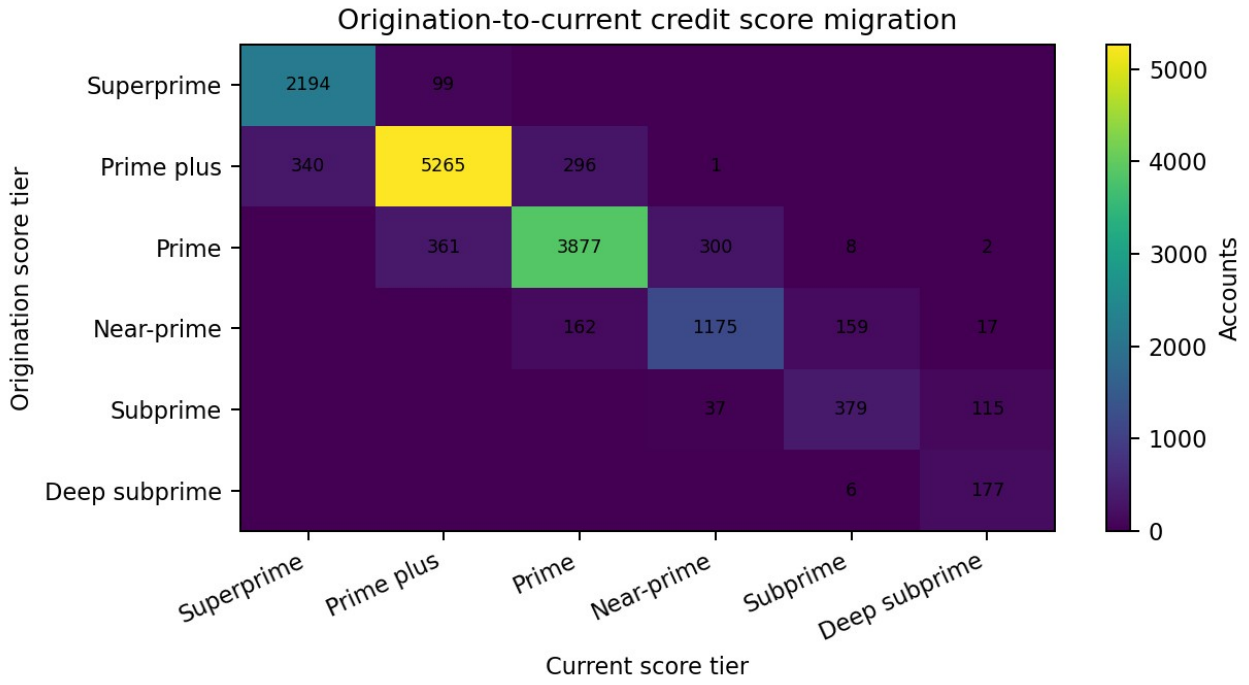
Key underwriting finding

Stage 2 is the central risk-management problem: 5.01% of exposure generates 47.53% of ECL. This supports account-level watchlist treatment rather than broad portfolio retrenchment.

05 | MIGRATION

Credit Score Migration & Deterioration

Movement from origination credit quality to the reporting-date score tier



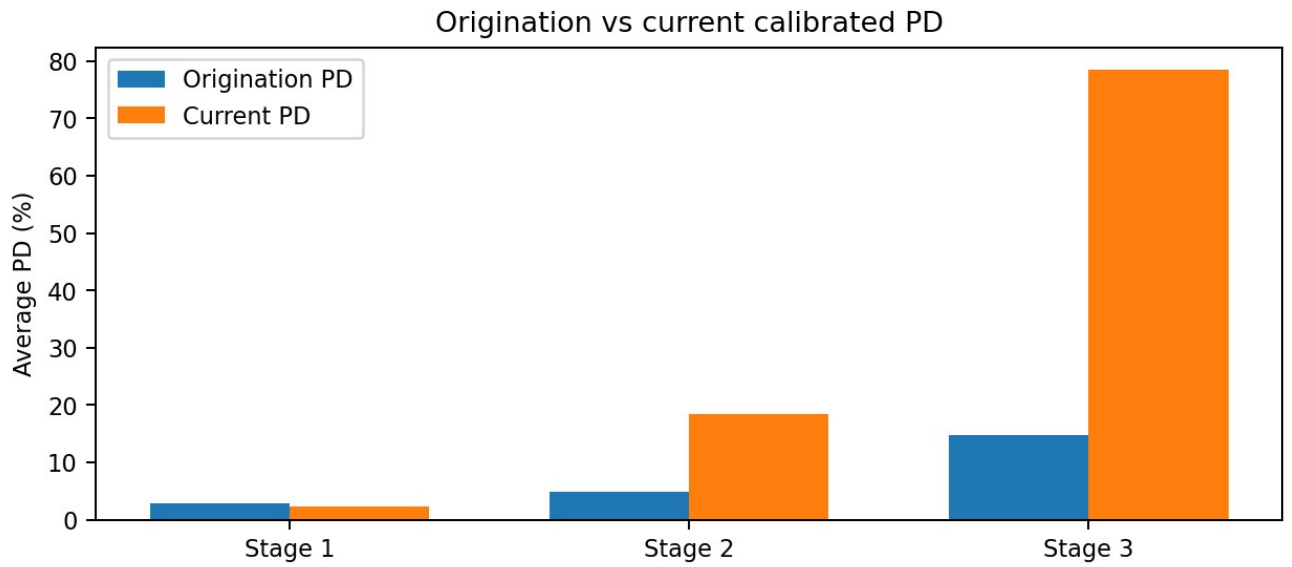
Interpretation

Diagonal cells indicate relative stability; movement toward worse current tiers signals deterioration. Near-prime and prime originations account for meaningful downgrade flows into Subprime / Deep subprime segments, which are disproportionately relevant to Stage 2 and Stage 3 loss.

Underwriting note: Score tiers are synthetic portfolio constructs used for risk segmentation; they are not external bureau ratings.

06 | PD UNDERWRITING

PIT Probability of Default - Model & Credit Implications



Stage	Avg origination PD	Avg current PD	Avg PD ratio	Underwriting implication
Stage 1	2.93%	2.26%	0.75x	Performing population is broadly stable/improved versus origination.
Stage 2	4.88%	18.46%	7.55x	Material deterioration supports lifetime-ECL treatment and watchlist attention.
Stage 3	14.67%	78.48%	21.26x	Credit-impaired population carries near-default / default-like current risk.

Model design

The PD champion is a 12-month PIT logistic model using borrower, delinquency, utilization, affordability, balance, tenure, product, and macro features. The target follows X_t -> default during t+1...t+12 to avoid look-ahead leakage.

Underwriting note: The negative unemployment coefficient observed in the PIT model is not used as a causal stress lever; scenario stress is applied through a separately governed overlay.

07 | PD VALIDATION

PD Validation, Calibration & Model Approval

0.909	0.901	0.897	0.0294
TRAIN AUC	VALID AUC	TEST AUC	TEST Brier

Metric	TRAIN	VALID	TEST	Credit-risk interpretation
AUC	0.9089	0.9012	0.8972	Strong and stable rank ordering across time.
KS	0.6615	0.6426	0.6331	Expected OOT softening without model collapse.
Observed default rate	5.04%	4.26%	4.31%	Risk environment improves versus TRAIN but remains above raw model mean.
Raw mean PD	5.04%	3.74%	3.42%	Raw OOT model underpredicts absolute risk.
Calibrated mean PD	-	-	3.89%	VALID-only calibration narrows the TEST gap.
PSI	0.000	0.042	0.083	Population drift remains below high-instability territory; continue monitoring.

Approval view

APPROVED / FROZEN. The champion retains strong OOT discrimination, improves calibration through VALID-only adjustment, and avoids TEST tuning. Remaining underprediction is documented rather than tuned away.

Underwriting note: Repeated account-month rows mean the temporal TEST is leakage-safe but not an independent borrower-level holdout.

08 | LGD UNDERWRITING

Loss Given Default, Collateral & Recovery Economics

27.2%	94.7%	86.4%	0.0688
Auto realized LGD	Card realized LGD	Personal realized LGD	Overall TEST RMSE

Product	Loss structure	TEST observed LGD	TEST predicted LGD	TEST RMSE	Underwriting interpretation
Auto Loan	Secured / collateral-sensitive	25.53%	21.10%	0.0857	Recovery value materially mitigates loss; collateral support must remain in model support.
Credit Card	Unsecured / low recovery	94.45%	94.98%	0.0434	Default avoidance and line management dominate recovery mitigation.
Personal Loan	Unsecured / partial recovery	85.74%	87.04%	0.1025	Severity remains high even when PD is moderate.

LGD architecture

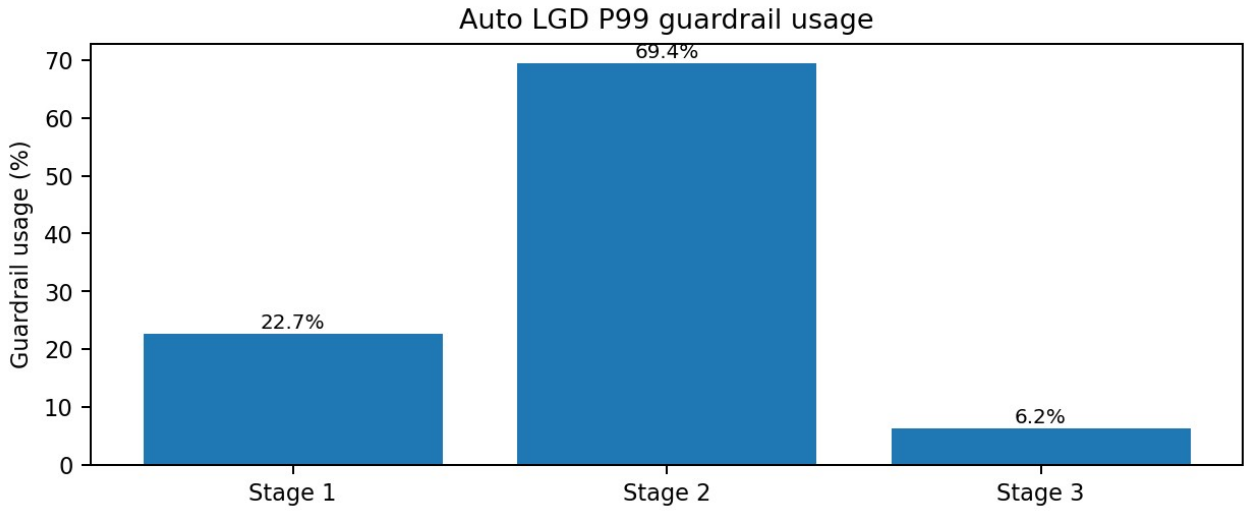
Auto uses secured severity modeling with collateral coverage and borrower/exposure features. Credit Card and Personal Loan use a hurdle framework: probability of any recovery plus conditional recovery severity.

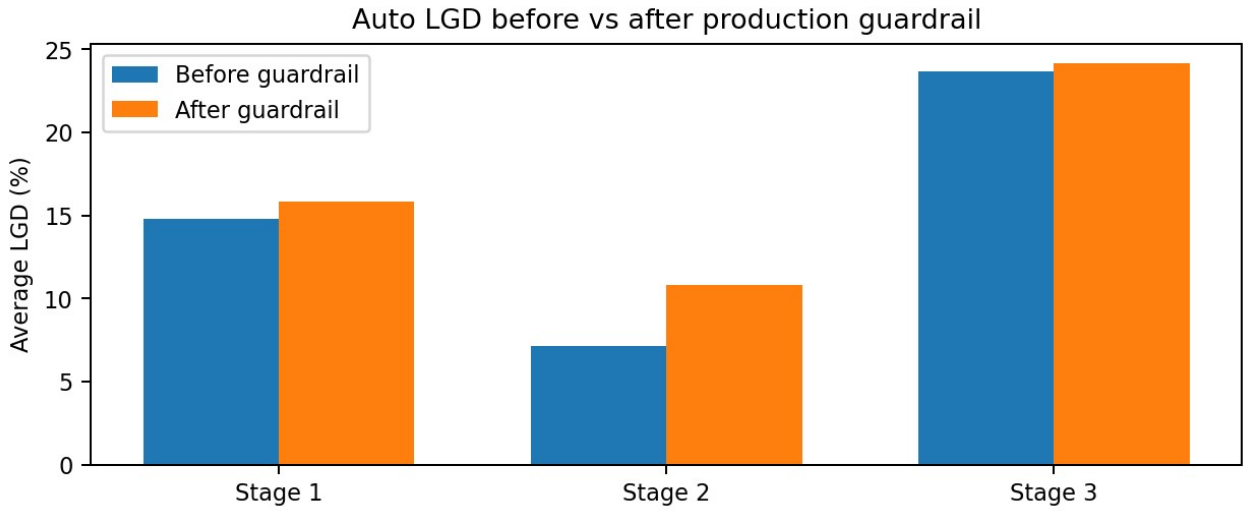
Underwriting note: Mature defaults are defined using an 18-month recovery horizon to avoid labeling incomplete workouts as final loss outcomes.

09 | MODEL RISK CASE STUDY

Auto LGD Production Guardrail

A production support issue detected after model validation





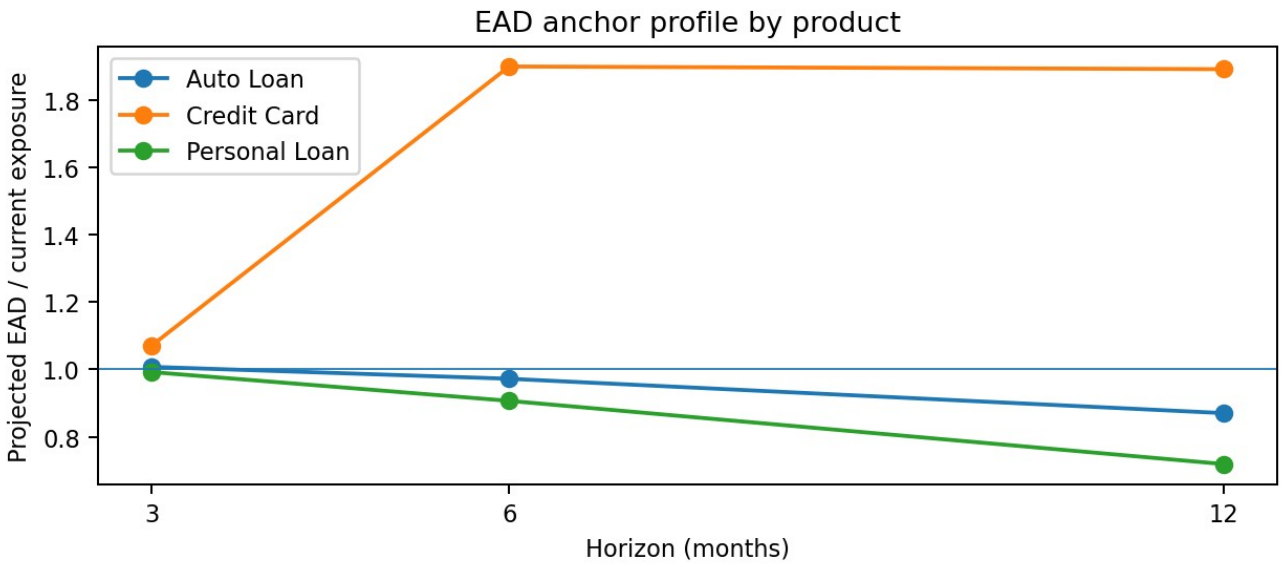
Why the control was required

TRAIN Auto default collateral coverage had P99 = 1.9894 and max = 2.0752, while production performing accounts often scored far beyond that support. Stage 2 was most exposed: 69.42% of Auto Stage 2 accounts required the guardrail.

Underwriting note: The model was not refit. Production scoring caps only the model input at TRAIN P99, while retaining raw collateral coverage for audit. Total portfolio ECL changed by only ~0.11%, consistent with a targeted support control rather than allowance manipulation.

10 | EAD UNDERWRITING

Exposure at Default / Credit Conversion Risk



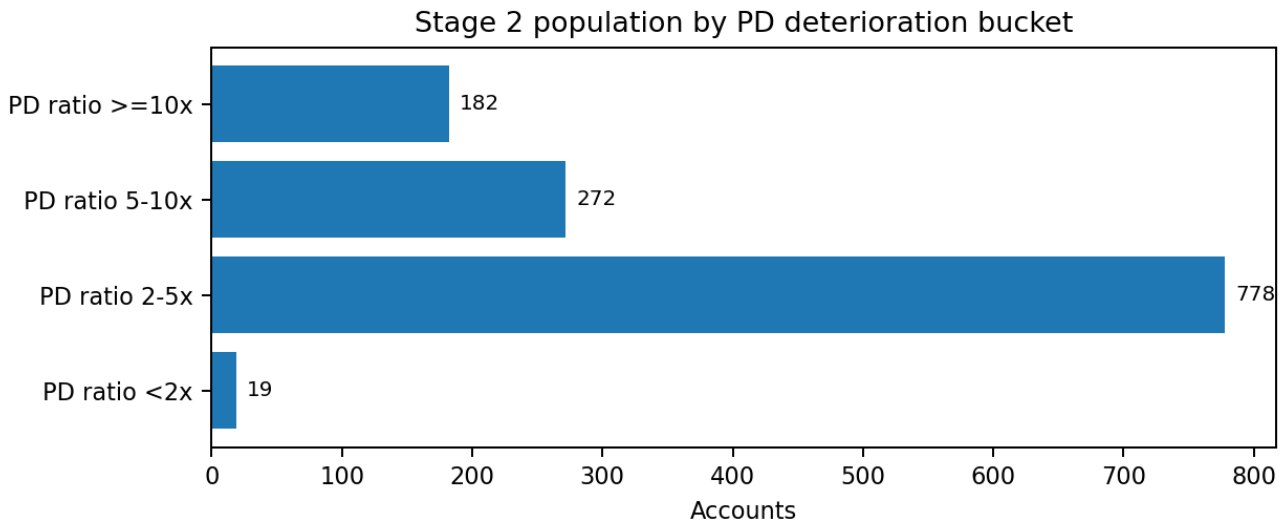
Product	3M EAD / current	6M EAD / current	12M EAD / current	Underwriting implication
Auto Loan	1.007x	0.972x	0.870x	Amortizing exposure declines by 12M.
Credit Card	1.071x	1.899x	1.891x	Undrawn line conversion creates EAD above current balance.
Personal Loan	0.992x	0.906x	0.719x	Scheduled amortization reduces conditional exposure.

Governance outcome

The first long-horizon EAD specification was rejected after unstable holdout behavior. The repaired architecture passed 9 of 9 product-horizon governance checks using a parsimonious Card 3M CCF model and stable development anchors for longer horizons.

11 | SICR POLICY

Significant Increase in Credit Risk & Stage 2 Drivers



Stage 2 trigger	Accounts	Share of Stage 2	Underwriting interpretation
PD deterioration >= SICR multiple	769	61.5%	Risk migration without 30+ DPD; requires early-warning response.
30 DPD + PD deterioration	463	37.0%	Both behavioral and modeled deterioration; highest-priority watchlist segment.
30 DPD backstop only	19	1.5%	Delinquency trigger despite PD ratio below the deterioration threshold.

Policy hierarchy

Stage 3 is assigned first for current default / credit-impaired status. Stage 2 is assigned for 30+ DPD or relative PD deterioration. Remaining accounts are Stage 1. Low-risk diagnostics do not override the conservative staging result.

Underwriting note: The project uses a relative 12-month PD multiple as a transparent SICR proxy. A bank implementation would typically compare lifetime risk over remaining maturity and incorporate institution-specific qualitative indicators and rebuttals.

12 | WATCHLIST CASE

Illustrative High-Risk Account Underwriting Snapshot

Highest Stage 2 weighted ECL account in the synthetic reporting portfolio

A000000162	Personal Loan	$28,066	$15,060
Account ID	Product	Exposure	Weighted ECL

Underwriting field	Observed value	Credit interpretation
Stage / reason	Stage 2 - SICR | 30 DPD backstop + PD deterioration	SICR is supported by both delinquency and modeled deterioration.
Current vs origination score	573 vs 598	Score deterioration of 25 points.
Delinquency	30 DPD	Behavioral evidence of heightened repayment risk.
Current calibrated PD	60.81%	High current 12M default likelihood.
Origination calibrated PD	24.52%	Meaningful deterioration from initial risk.
Raw PD ratio	2.61x	Exceeds the project SICR multiple.
Production LGD	85.94%	Unsecured severity is high if default occurs.
12M EAD	$20,175	Conditional exposure remains material through the risk horizon.
Downside ECL	$17,355	Loss increases materially under adverse conditions.

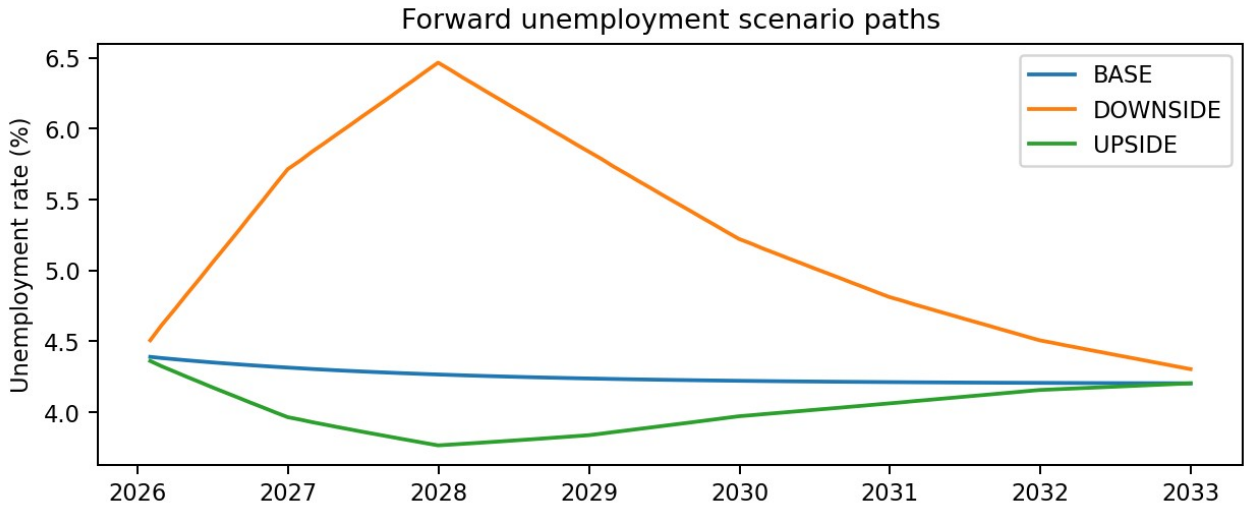
Illustrative account action

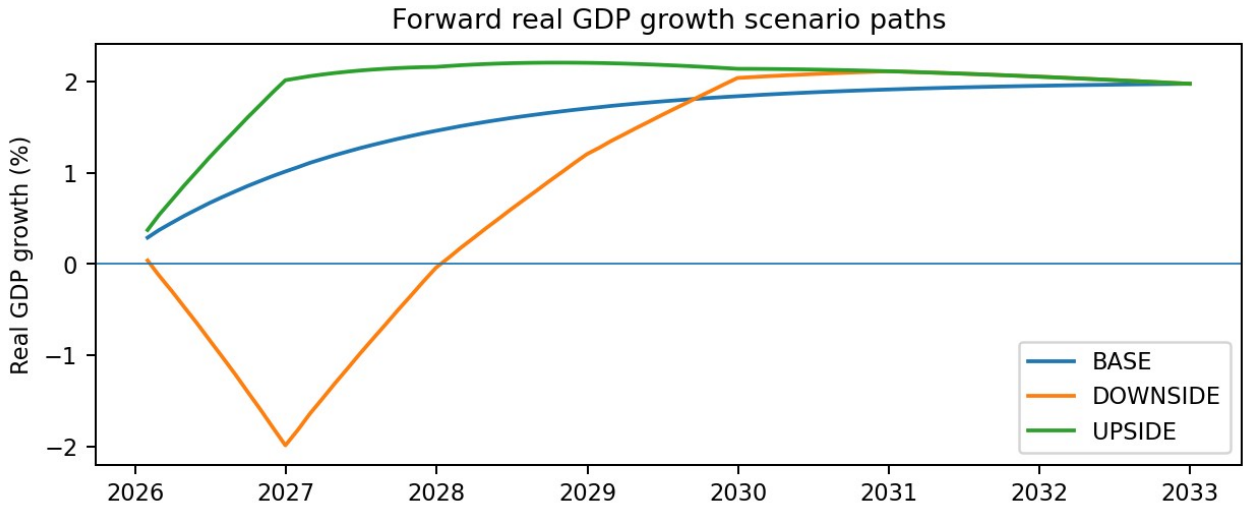
WATCHLIST / ENHANCED MONITORING. Prioritize repayment behavior, delinquency cure, exposure containment, and worsening PD trajectory. This synthetic account is presented to demonstrate account-level underwriting logic, not a live borrower decision.

13 | FORWARD-LOOKING RISK

Macroeconomic Scenario Framework

84-month Base / Upside / Downside paths with 60% / 20% / 20% probability weights





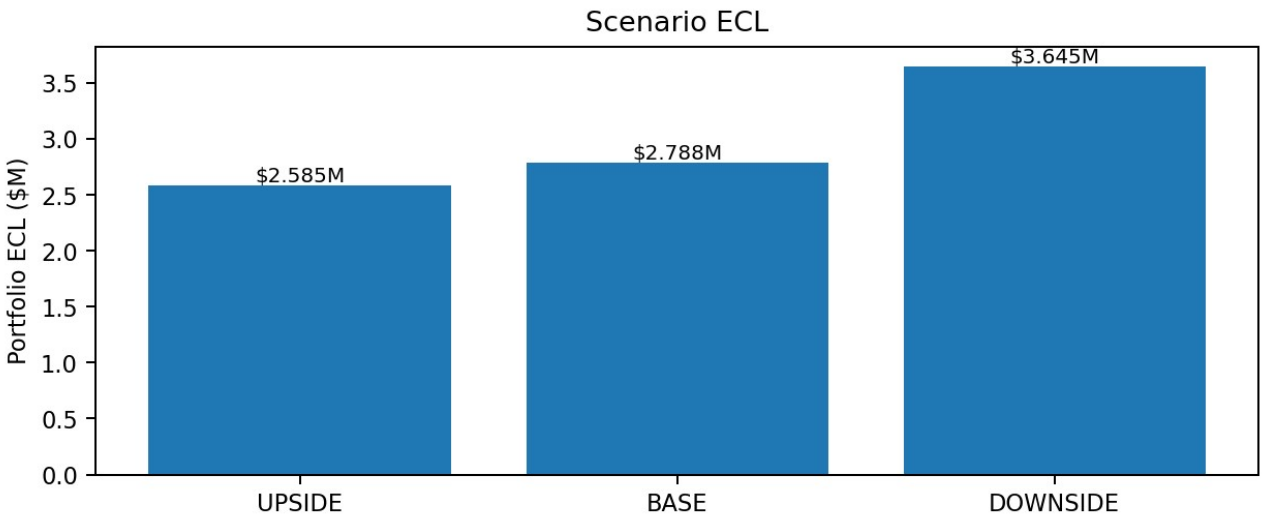
Scenario governance

GDP ordering is governed on cumulative GDP level rather than requiring monthly GDP growth to remain ordered every month. This avoids false failures when a downside economy rebounds faster in percentage growth but remains below the Base path in level terms.

Underwriting note: Scenario stress betas are transparent project assumptions used to translate macro stress into PD/LGD/EAD overlays; they are not causal coefficients estimated from the PIT PD model.

14 | STRESS TESTING

Scenario ECL & Product Sensitivity



Product / stage	Downside / Base	Risk interpretation
Auto - Stage 1	1.590x	Most macro-sensitive segment despite performing status.
Auto - Stage 2	1.372x	SICR exposure remains highly stress responsive.
Credit Card - Stage 1	1.383x	Unsecured portfolio sensitivity is material before default.
Credit Card - Stage 2	1.169x	Lifetime loss is already elevated, reducing incremental stress multiple.
Personal Loan - Stage 1	1.333x	Meaningful downside uplift with high unsecured severity.
Credit-impaired Card / Personal	~1.01x - 1.04x	Already severe/default-like risk leaves less incremental scenario room.

Stress conclusion

Downside ECL of $3.645M is $857K (+30.7%) above Base. The stress response is material but ordered and explainable, supporting portfolio retention with targeted sensitivity monitoring rather than an across-the-board exposure cut.

15 | ALLOWANCE

Lifetime ECL & Allowance Adequacy

$2.919M	2.255%	$2.557M	$361K
Weighted ECL	Allowance Rate	Weighted 12M ECL	Stage 2 Lifetime Uplift

Stage	Exposure	Weighted ECL	Coverage	Lifetime effect
Stage 1	$122.45M	$1.287M	1.051%	12-month ECL only.
Stage 2	$6.49M	$1.387M	21.387%	Lifetime ECL; +$361K over 12M treatment.
Stage 3	$0.52M	$0.245M	46.889%	Simplified immediate credit-impaired loss treatment.

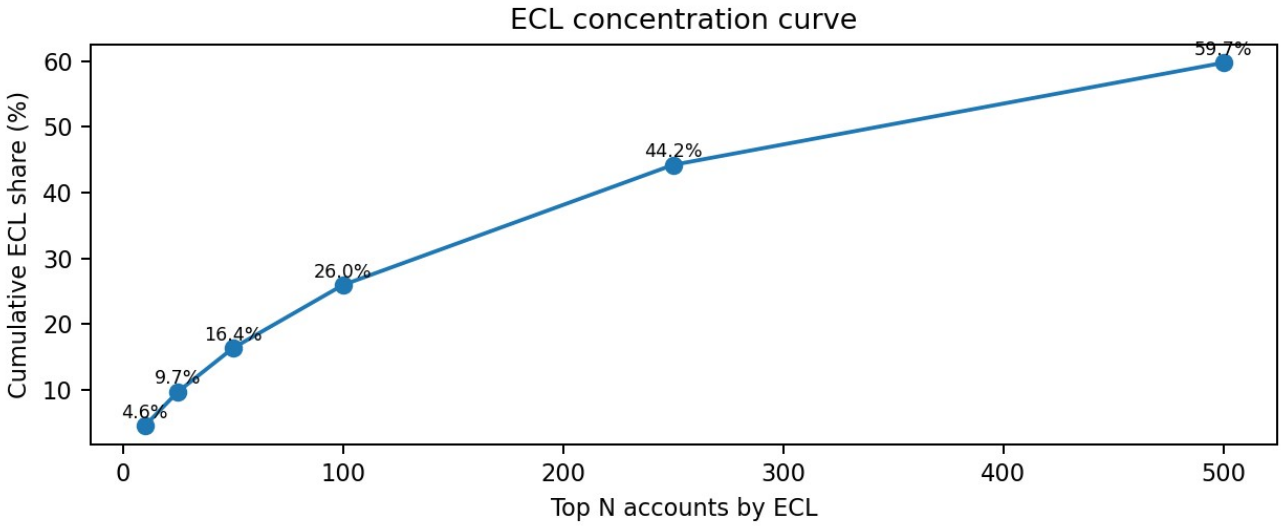
Stage 2 product	12M ECL	Lifetime ECL	Lifetime / 12M	Incremental lifetime ECL
Auto	$289K	$370K	1.277x	+$80K
Credit Card	$498K	$740K	1.486x	+$242K
Personal Loan	$238K	$278K	1.165x	+$39K

Allowance view

The allowance is risk-concentrated rather than balance-concentrated. Stage 2 lifetime treatment is economically meaningful, especially for revolving Credit Card exposures, and is a major driver of the portfolio reserve.

16 | CONCENTRATION

Expected Loss Concentration & Watchlist Prioritization



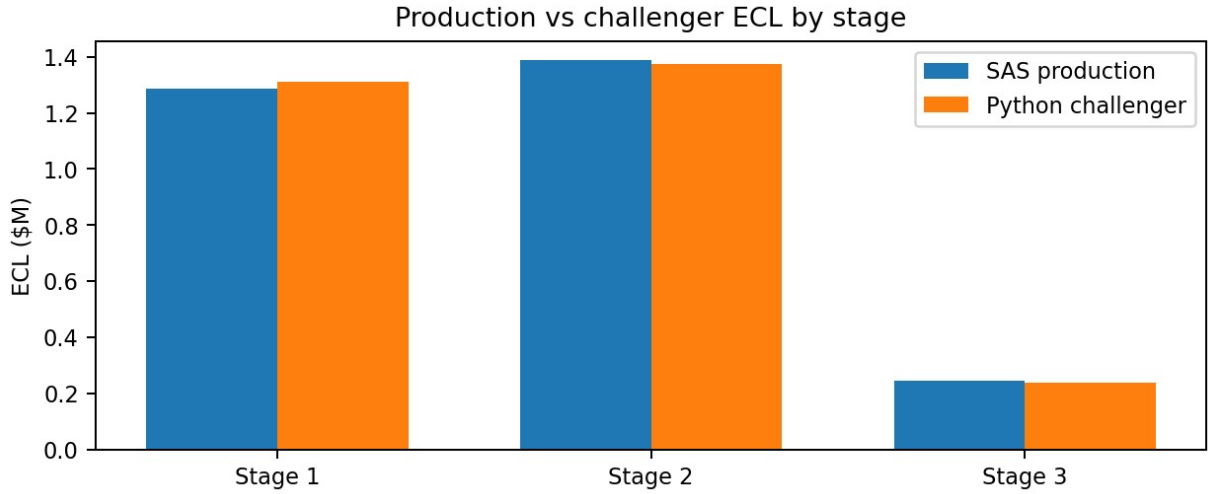
Top N accounts by ECL	Share of total ECL	Share of total exposure	Underwriting implication
10	4.56%	0.26%	Very high loss density in the largest individual risks.
25	9.73%	0.60%	Loss concentration materially exceeds balance concentration.
100	25.95%	1.50%	Quarter of allowance is concentrated in <1% of accounts.
250	44.20%	2.72%	Focused watchlist can cover almost half of ECL.
500	59.72%	4.75%	Top 3.34% of accounts drive nearly 60% of ECL.

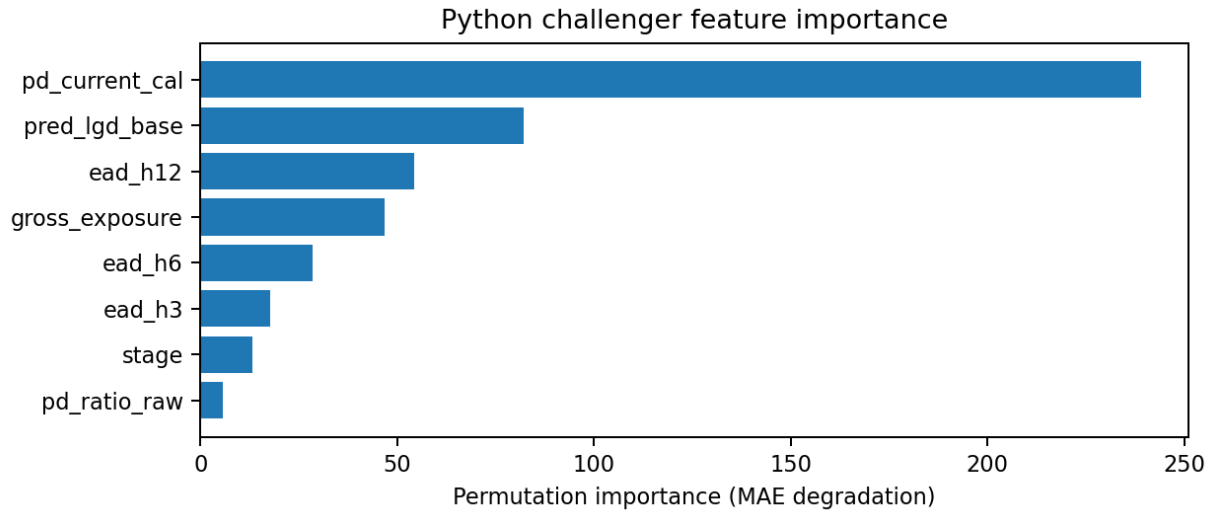
Credit-management implication

Prioritize risk review by expected loss, not by exposure alone. A Top-500 ECL watchlist covers nearly 60% of the allowance while representing only 4.75% of exposure.

17 | GOVERNANCE

Model Validation, Python Challenger & Control Environment





97.81%	+0.17%	3 / 3	5 / 5
Challenger R2	Aggregate Difference	Reporting Gates	ECL Gates

Governance conclusion

SAS remains the production system of record. The Python model is an independent surrogate/explainability layer; it does not replace or retune production. Feature importance aligns with expected ECL economics: PD, LGD, EAD, exposure, stage, and deterioration dominate.

18 | ONGOING MONITORING

Recommended Monitoring Framework & Escalation Triggers

Illustrative governance thresholds for portfolio management - not regulatory standards

Risk dimension	Current position	Illustrative amber trigger	Illustrative red trigger	Recommended action
Stage 2 exposure share	5.01%	>7.5%	>10.0%	Monthly migration review; tighten new origination / line actions by product.
Portfolio allowance rate	2.255%	>2.75%	>3.50%	Reforecast ECL drivers and scenario contribution.
Downside / Base ECL	1.307x	>1.40x	>1.60x	Escalate stress sensitivity and management overlay review.
Auto guardrail usage	26.19%	>30%	>40%	Review model support, collateral feature distribution, and possible redevelopment.
Stage 2 Auto guardrail usage	69.42%	>75%	>85%	Prioritize secured-LGD support analysis.
PD TEST PSI	0.083	>0.10	>0.25	Investigate population drift; recalibration / redevelopment assessment.
Top-500 ECL concentration	59.72%	>65%	>75%	Tighten watchlist concentration controls and case review cadence.

Recommended cadence

- Monthly: Stage migration, delinquency, PD drift, guardrail usage, and Top-ECL watchlist.
- Quarterly: scenario sensitivity, product allowance, challenger reconciliation, and model-support review.
- Semiannual / annual: calibration, stability, EAD/LGD backtesting, assumptions review, and redevelopment decision.

Escalation principle

Monitoring should be action-linked. Threshold breaches should trigger a defined credit or model-risk response, not merely a dashboard alert.

19 | FINAL COMMITTEE VIEW

Final Underwriting Conclusion & Key Limitations

Final portfolio conclusion

ACCEPTABLE / MAINTAIN WITH ENHANCED MONITORING. The portfolio demonstrates strong model governance and a predominantly performing exposure base, but loss is concentrated in Stage 2, unsecured severity is structurally high, and selected Auto LGD production observations require an explicit support guardrail.

Committee consideration	Conclusion
Credit quality	94.59% of exposure remains Stage 1; deterioration is concentrated rather than portfolio-wide.
Expected loss	$2.919M probability-weighted ECL / 2.255% portfolio coverage.
Stress resilience	Downside raises ECL by $857K / 30.7% versus Base; response remains economically ordered.
Model performance	PD OOT AUC 0.897; LGD and EAD champions pass designated validation tests.
Production control	Auto LGD TRAIN-P99 guardrail addresses out-of-support scoring without refitting to final ECL.
Governance	5/5 final ECL gates and 3/3 reporting reconciliation gates pass.

Known limitations retained for committee transparency

- Synthetic loan-level portfolio; no real bank customer or proprietary borrower data.
- Scenario overlay betas are governed assumptions, not causal macro elasticities.
- Stage 3 reflects reporting-date visible/default population rather than a reconstructed workout-stock ledger.
- Stage 3 ECL uses simplified immediate-loss mechanics instead of a full discounted workout cash-shortfall model.
- Stage 2 revolving EAD after 12 months uses a controlled behavioral-life assumption.
- Python challenger R2 measures explanatory fidelity to production ECL, not realized-loss predictive accuracy.

Sign-off

Prepared as a portfolio underwriting / credit-risk work sample. The decision language is illustrative and demonstrates committee-style risk communication; it is not legal, accounting, regulatory, or investment advice.

Suggested committee actions: Maintain portfolio; preserve current allowance framework; continue Stage 2 watchlist escalation; monitor Auto guardrail usage; retain explicit Stage 3 limitation until a workout-stock bridge is implemented.